

Research

Economic burden and quality of life of maintenance hemodialysis patients in a rural area of South India: a pharmacoeconomic evaluation

Redlin Jani Rajan¹ · Reema Mohammed Sultan¹ · Sanjana Mariam Saju¹ · Panneerselvam Theivendren² · Nithya Raju¹

Received: 31 August 2024 / Accepted: 28 April 2025

Published online: 17 May 2025

© The Author(s) 2025 **OPEN**

Abstract

Objective The intersection of economics and quality of life in hemodialysis patients with chronic kidney disease–End-stage renal disease is complex. Hemodialysis is a life-saving treatment for patients with end-stage kidney disease, but it is also a very expensive treatment and has a considerable influence on a patient's quality of life, both physically and emotionally. This study aims to predict the cost of illness and quality of life in maintenance dialysis patients in a rural area of South India.

Methods Individuals from tertiary care hospitals (N=55) were filtered to satisfy the study inclusion criteria. We used the cost of illness as a predictor to analyse patients' direct cost, indirect cost and intangible cost using the Kidney Disease Quality of Life 36 for a period of 6 months. Costs were estimated in terms of Indian national rupees (INR; 1 USD ~ INR82.52).

Results In our study, 76.3% of patients underwent haemodialysis thrice a week. A total of 52.7% of patients spent INR 100–200 (USD 1.20–2.40) on laboratory investigations monthly. Additionally, 47.2% of patients lost wages between INR 1000–3000 (USD 12–36) per visit, and 52.7% of patient caregivers lost less than INR 3000 (USD 36) due to this procedure. The Kidney Disease Quality of Life-36 analysis revealed a significant impact ($p < 0.05$) on physical and mental functioning, the burden of kidney disease, symptoms, and the effects of kidney disease on daily life in haemodialysis patients.

Conclusion The cost of maintenance haemodialysis like the direct, indirect, and intangible costs significantly burdens patients and caregivers, highlighting the need for additional support and care, particularly those from lower socio-economic backgrounds in rural areas. This study emphasizes the need for improved access to hemodialysis treatment and support services for critical review.

Keywords Cost of illness · Hemodialysis · Economic burden · Quality of life · Rural South India

Redlin Jani Rajan, Reema Mohammed Sultan, Sanjana Mariam Saju and Panneerselvam Theivendren contributed equally to this work.

✉ Nithya Raju, nithyapharma14@gmail.com; Redlin Jani Rajan, redlinjani729@gmail.com; Reema Mohammed Sultan, msreema153@gmail.com; Sanjana Mariam Saju, sanjana0saju@gmail.com; Panneerselvam Theivendren, tpsphc@gmail.com | ¹Department of Pharmacy Practice, Swamy Vivekanandha College of Pharmacy, Namakkal, Tamil Nadu 637205, India. ²Department of Pharmaceutical Chemistry, Swamy Vivekanandha College of Pharmacy, Namakkal, Tamil Nadu 637205, India.



1 Introduction

Chronic kidney disease (CKD) and end-stage renal disease (ESRD) are significant public health challenges in India, affecting millions of people. According to the Indian Society of Nephrology, approximately 175 million people suffer from CKD, and over 3 million have End-stage renal disease [1]. This number is projected to rise in the coming years. Hemodialysis is a critical treatment for Chronic kidney disease–End-stage renal disease, as it removes waste products and excess fluid from the blood, improving patients' health and quality of life. Additionally, it helps prevent complications such as heart disease, stroke, and anemia [1–5]. However, access to hemodialysis in India is limited, with only around 10% of Chronic kidney disease–End-stage renal disease patients receiving this life-saving treatment [7]. The cost of hemodialysis is a major concern in India, contributing to the poor quality of life and higher mortality rates among Chronic kidney disease–End-stage renal disease patients [6]. The average annual cost of treatment is estimated to be INR 1.4 lakhs (USD 1800), which is a significant financial burden, especially for low-income households [8]. The high costs are driven by factors such as expensive dialysis machines and supplies, medical staff salaries, drugs to manage the condition, transportation, and loss of income due to the inability to work [10, 11]. Many patients resort to debt or selling assets to afford treatment, which leads to financial hardship, stress, and anxiety. India's healthcare system does not provide universal health coverage, and the government's healthcare spending is low at only 4% of GDP, far below the global average [9]. Most healthcare costs are borne by individuals and families, further exacerbating the financial burden on Chronic kidney disease. Although the government has introduced programs to provide free hemodialysis to poor patients, these initiatives are not widely available and are insufficient to meet the growing demand [12]. The availability of hemodialysis centers in rural areas of South India is particularly limited compared to urban areas. This disparity creates additional challenges for rural patients, who often have to travel long distances for treatment, incurring higher transportation costs. Rural patients also tend to have lower incomes, making it even more difficult for them to afford the already expensive treatment [13]. There is a lack of comprehensive data on the cost of illness as presented in Fig. 1 of Chronic kidney disease in rural South Indian populations due to the absence of a national registry and limited studies on this subject. Therefore, this study aims to predict

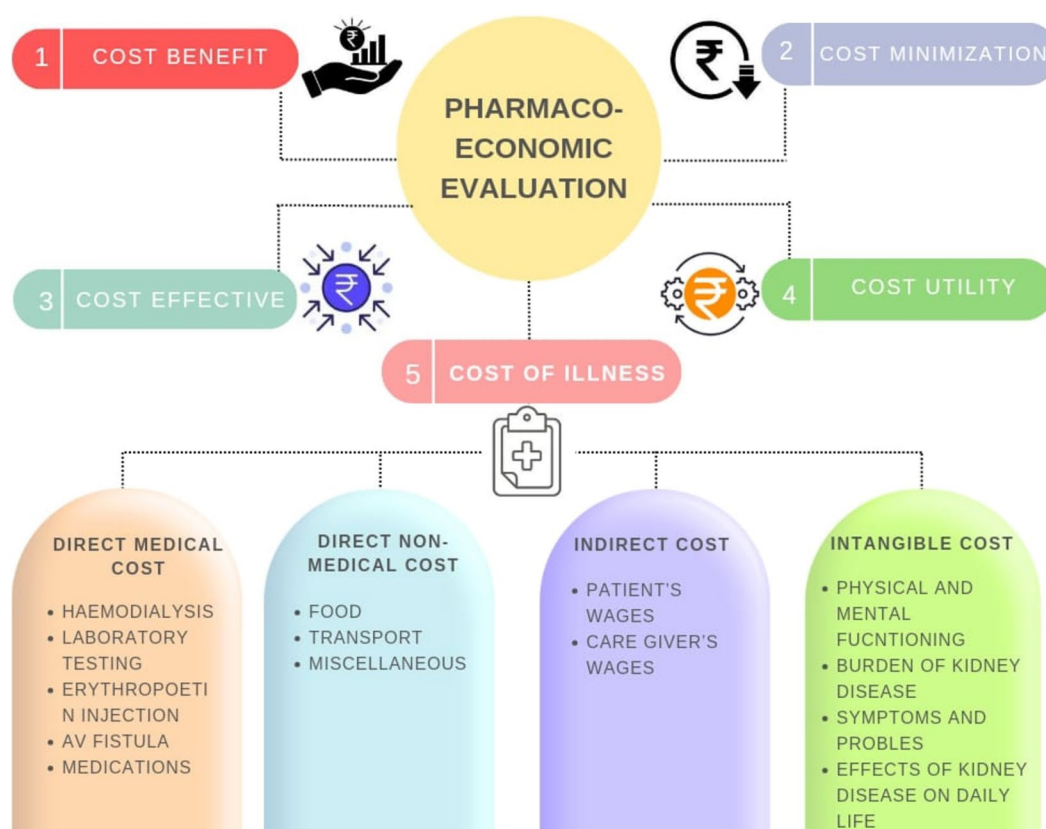


Fig. 1 Classification of cost of illness

the cost of illness and assess the quality of life among maintenance dialysis patients in rural South India. The findings of this study could help inform policy decisions and improve access to affordable dialysis treatment in underserved areas.

2 Methods

2.1 Study setting and study design

This is a prevalence-based prospective observational study focusing on cost of illness (in INR) and quality of life (Kidney Disease Quality of Life 36) from a patient perspective. In economic evaluations of healthcare interventions, it is essential to consider the patient perspective to ensure that the analysis reflects the impact of the intervention on patient well-being and outcomes. This investigation was conducted over a period of 6 months at a tertiary care hospital located in a rural area within the southern region of India. This study was conducted in accordance with the ethical standards of the institutional and national research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. Ethical approval was obtained from the Institute Ethics Committee of Vivekanandha Medical Care Hospital, File Ref. No: EC/NEW/INIT/2021/1811 reviewed and discussed to conduct the study. (Study Ref. no: SVCP/IEC/MAR/2023/08). The participants are fully aware of this study and had signed the Informed consent.

2.2 Study population

We employed purposive sampling, a nonprobability sampling technique, to deliberately select a total of 60 participants. The primary aim was to assess the generalizability of the findings related to the cost of illness and to investigate the impact of our selection criteria. Eventually, we narrowed our selection down to 55 individuals from the larger population as three patients had deceased during the study period and two were unwilling to participate in the study.

Our selection criteria encompassed both male and female patients who had been receiving maintenance haemodialysis for a minimum of 3 months, whether or not they had concurrent health conditions. We included both patients who were already undergoing haemodialysis at the study's outset, referred to as prevalent patients, and those who were newly enrolled in the study, referred to as consecutive patients. Patients were excluded from participation if they were undergoing haemodialysis for reasons other than Chronic kidney disease–End-stage renal disease, had a planned renal transplant scheduled within the study period, declined to be part of the study, or were unable to complete the Kidney Disease Quality of Life-36 questionnaire. Consent was acquired from the adult individuals undergoing maintenance haemodialysis after providing them with relevant information regarding the investigation. To mitigate selection bias, the study defined clear inclusion/exclusion criteria and used the hospital's dialysis registry for consistent recruitment, ensuring every eligible patient had an equal chance of inclusion. Efforts to minimize non-response bias included thorough participant engagement and follow-ups, while anonymity and standardized data collection ensured accurate, unbiased responses. Each participant was assigned a numerical code for identification purposes, ensuring the questionnaire's anonymity. The obtained results were kept confidential and accessible only to the researcher and the supervisors.

2.3 Data collection

The research approach embraced a patient standpoint, encompassing the entirety of costs, regardless of their source. The cost of illness analysis was structured into three essential stages: first, the identification of costs, quantification, and finally, the assignment of a valuation.

Step 1: The identification of costs

The data collection process involved gathering information from case sheets and conducting direct interviews with both patients and their caregivers. We utilized a specially designed data entry form for this purpose. The collected data encompassed various categories, including demographic details, direct costs, indirect costs, and intangible costs, all of which were assessed using the Kidney Disease Quality of Life -36 questionnaire as represented in Fig. 2. Additionally, we analysed the socioeconomic status of the 55 participants by employing Prasad's modification of classification as represented in Table 1, which is based on per capita income [14].

Step 2: Quantification

Fig. 2 Intangible cost-Domains of KDQOL-36

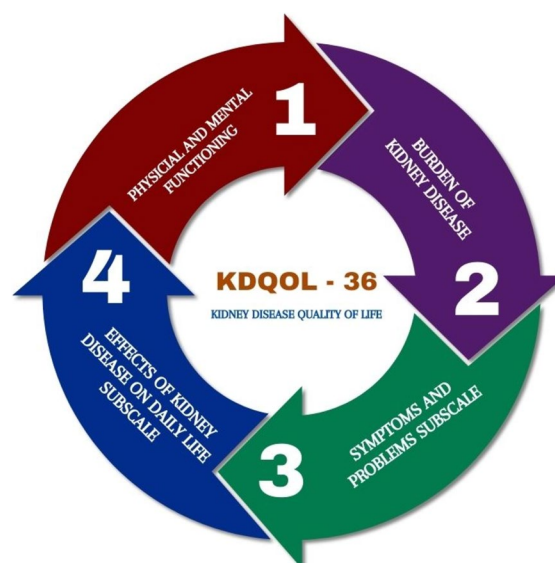


Table 1 Modified by Prasad's classification based on per capita income

Per capita income (in INR/month)	Category
> 7008	Upper class
3504–7007	Upper-middle class
2102–3503	Middle class
1051–2101	Lower middle class
< 1050	Lower class

In the category of Direct Cost, we considered various components, including the expenses associated with Haemodialysis, Laboratory Investigation, Injection Erythropoietin, AV Fistula, and Medications. We included the cost of the specific brand of medication that was prescribed. If drugs were prescribed by their generic name, we took into account the highest cost associated with those generics. The annual cost of treatment was then calculated based on these factors. Additionally, under Direct Cost, we factored in expenditures related to food and transportation, which fall under direct nonmedical expenses. For indirect cost data, we gathered information regarding patient wages, caregiver wages, and other related costs through interviews with the patients and by reviewing bills. Again, the annual cost of treatment was determined based on these details. Furthermore, to capture Intangible Costs, we conducted interviews with patients using the Kidney Disease Quality of Life TM (KDQOL TM-36) version 1. This questionnaire, copyrighted in 2000 by the RAND Corporation, is a specialized tool designed to assess the quality of life specific to the kidney disease. It covers aspects such as symptoms, the burden of illness, social interactions, staff encouragement, and patient satisfaction. The Kidney Disease Quality of Life -36 consists of 36 items and comprises four distinct subscales aimed at measuring health-related quality of life (HRQOL).

The Kidney Disease Quality of Life-36 questionnaire covers four key domains to assess the quality of life in kidney disease patients. The "Physical and Mental Functioning" domain evaluates the patient's overall health, daily activities, mood, energy levels, and social engagement. "Burden of Kidney Disease" measures how kidney disease impacts daily life, time spent managing it, frustration levels, and feelings of burden. The "Symptoms and Problems" domain assesses discomfort from symptoms such as sore muscles, chest pain, and issues with dialysis access. Last, "Effects of Kidney Disease on Daily Life" explores its impact on fluid and diet restrictions, work, travel, reliance on medical staff, stress, sex life, and personal appearance. These domains provide a comprehensive view of a patient's experience, guiding tailored care and support.

Step 3: The assignment of a valuation

The costs were valued by using data from the questionnaire as well as preexisting cost information, which included expenses related to items such as medication and outpatient visits. All calculations were reported in Indian rupees. When there was a lack of cost information, the approach was to use the minimum cost estimate. Furthermore, in cases where

participants reported undergoing maintenance hemodialysis more frequently than three times a week, the calculations were conducted based on a maximum of three cycles per week. The computation of productivity loss for both the study population and their caretaker is equivalent to the amount the employer would have disbursed if the individual had been actively working. This involves multiplying the total duration of absenteeism by the daily cost.

2.4 Data entry and analysis

In our analysis, Categorical variables were presented as frequencies and percentages, while continuous variables were expressed as means $\pm$ standard deviations (SD). In cases where the distribution of continuous variables did not follow normality, appropriate non-parametric tests were utilized. To compare differences across groups within the patient, we employed the Kruskal–Wallis test. This test was used to evaluate differences Kidney Disease Quality of Life (KDQOL-36) subscales applied between pre and post patients undergoing hemodialysis. P-values from the Kruskal–Wallis test indicate whether these differences were statistically significant at a significance level of $p < 0.05$. It is important to note that these comparisons were conducted within the patient population before and after hemodialysis, rather than in comparison to a healthy population.

All statistical analyses were conducted using IBM SPSS Statistics version 22.0. Additionally, for clarity, we calculated the cost per month for individual components like Erythropoietin, IV Iron, food, transport, and wages to standardize the costs. Furthermore, all financial values were reported in INR and USD (converted at the prevailing exchange rate during the study period) to cater to an international readership as represented in Fig. 3.

3 Results

The subgroup analysis of maintenance hemodialysis patients was collected and categorized, and the results are presented in Table 2.

The majority of maintenance hemodialysis patients in the study groups were middle-aged males with low socioeconomic status. They had a high incidence of comorbidities such as diabetes mellitus and hypertension. The direct and indirect costs of haemodialysis were significant, and many patients were unable to afford them. This study found that the majority of patients were in the 51–60 age group (81.8%) and were male (81.8%). The majority of male patients were

Fig. 3 Methodology for the analysis of economics and quality of life in hemodialysis patients with CKD-ESRD

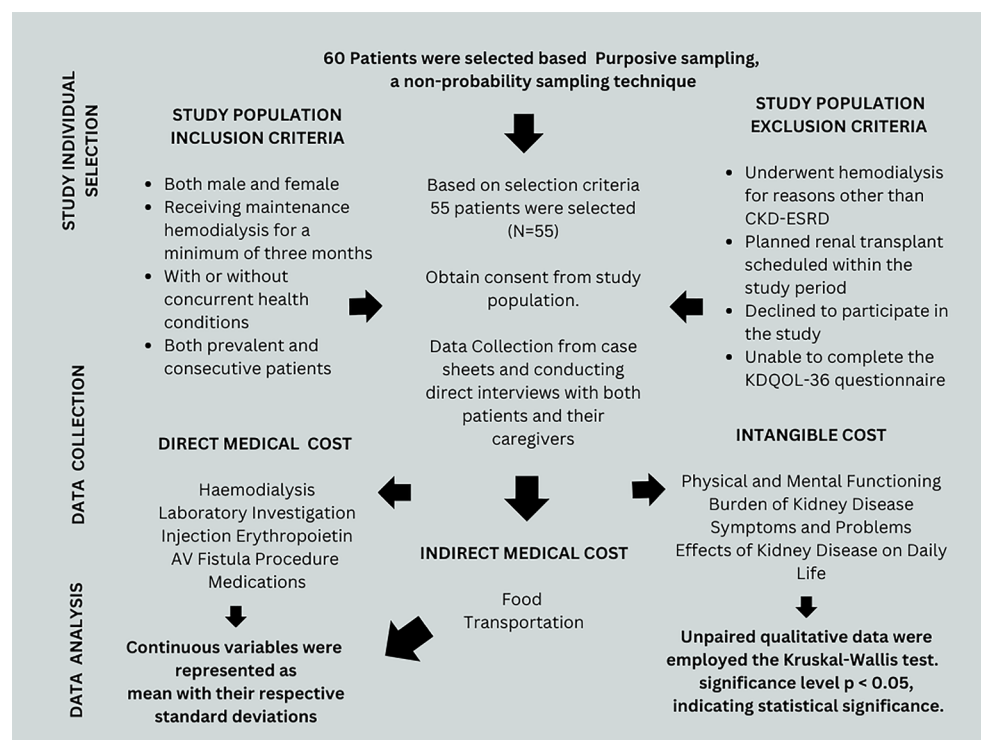


Table 2 Subgroup analysis of hemodialysis patients (N = 55)

Category	Number of patients(N = 55)	Percentage
Age (in years)		
31–41	6	10.9
41–50	8	14.5
51–60	18	32.7
61–70	16	29.1
71–80	7	12.7
Gender		
Male	46	81.8
Female	9	18.2
Social history		
Male (n = 46)		
Smoker	20	43.4
Alcoholic	11	23.9
Tobacco	5	10.8
All the above	6	13
None	4	8.6
Female (n = 9)		
Tobacco	3	33.3
None	6	66.6
Comorbidities		
DM,HTN	3	5.5
CKD	7	12.7
DM,HTN,DM	17	30.9
HTN	7	12.7
CKD-ESRD	16	29.1
DM,CKD	1	1.8
Hepatitis C, HTN	2	3.6
Hepatitis C, DM, HTN	2	3.6
Family history with kidney disease		
Yes	12	21.8
No	43	78.2
BMI		
Normal (18.5–24.9)	36	65.5
Underweight (< 18.5)	2	3.6
Overweight (25.0–29.9)	12	21.8
Obese (> = 30)	5	9.1
Duration (in years)		
0–1	20	36.4
1–2	18	32.7
2–3	4	7.3
Above 3	13	23.6
Educational status		
Illiterate	36	65.5
School	10	18.2
Degree	9	16.3
Monthly income		
> 7008	2	3.6
3504–7007	3	5.4
2102–3503	23	41.8
1051–2101	17	30.9
> 1050	10	18.1
Marital status		
Married	46	83.6
Divorced	2	3.6
Single	7	12.8

Table 2 (continued)

DM: Diabetes Mellitus; HTN: Hypertension; CKD: Chronic Kidney Disease; CKD-ESRD: Chronic Kidney Disease with End Stage Renal Disease; BMI: Body Mass Index; OBESE: Obesity

also smokers, alcoholics, and tobacco users (91.1%). The study population had a higher incidence of Diabetes mellitus and hypertension as comorbidities. Approximately 36.4% of the population had been on maintenance haemodialysis for the past year.

Modified by Prasad's classification for socioeconomic status, 41.8% of the patients had a daily income within INR 2102–3503 (USD 25–42). A total of 18.1% had an income under INR 1050 per day (USD 13), and 65.5% of the study population was illiterate. A total of 83.6% of the population was married. In direct medical costs, 76.3% of patients had their haemodialysis procedure thrice a week. A total of 52.7% of patients spent INR 100–200 (USD 1.20–2.40) on laboratory investigations monthly. A total of 67.2% spent INR 1300 (USD 16) for Inj. Erythropoietin, and 32.7% spent INR 300 (USD 3.60) for Inj. Iron Sucrose weekly. For AV fistula access, 54.5% spent INR 4550 (USD 55) for double vein access, while 45.4% spent INR 5950 (USD 72) at once. Additionally, 29% of the population spent INR 600–700 (USD 7–8.50) on medication monthly.

In direct nonmedical costs, 56.3% of the population spent INR 200–300 (USD 2.40–3.60) weekly on food, and 52.7% of the population spent less than INR 100 (USD 1.20) weekly on transport. A total of 47.2% of patients lost wages between INR 1000–3000 (USD 12–36) per visit, and 52.7% of patient caregivers lost less than INR 3000 (USD 36) in wages due to this procedure.

These findings suggest that the majority of maintenance haemodialysis patients in India face significant financial challenges. The high costs of direct medical care, nonmedical expenses, and lost wages place a considerable burden on patients and their families, as represented in Table 3. Patients undergoing hemodialysis incurred significant direct medical costs, with monthly expenses for two and three sessions being INR 16,000 (USD 192) and INR 24,000 (USD 288), respectively. Non-medical costs like food (INR 100–300 or USD 1.20–3.60) and transport (INR 1–200 or USD 0.12–2.40) added to the overall burden. Additionally, patients and caregivers faced substantial indirect costs due to wage losses, ranging from INR 1000 to over INR 3000 (USD 12–36) per visit, as represented in Table 4. It is important to develop policies and programs to support these patients and their families. The intangible costs of haemodialysis were analysed using the Kidney Disease Quality of Life-36 survey, which assesses the quality of life of patients with kidney disease. The survey includes six domains, including physical functioning, role-physical, bodily pain, general health, vitality, social functioning, role-emotional, mental health, sexual functioning, sleep problems, health distress, and burden of kidney disease.

The first domain of the Kidney Disease Quality of Life-36 contains information about the physical and mental functioning of patients undergoing maintenance haemodialysis. The study found that the impact of physical and mental functioning on haemodialysis patients is significant (p value < 0.005) as presented in Table 5.

The second domain of the Kidney Disease Quality of Life-36 survey assesses the burden of kidney disease in patients undergoing maintenance haemodialysis. The study found that the incidence of burden due to kidney disease was statistically significant (p value < 0.005) as shown in Table 6. The third domain in Kidney Disease Quality of Life-36 contains information regarding the symptoms and problems in patients undergoing maintenance haemodialysis. This study result indicates statistical significance (p value < 0.005) with symptoms such as muscle soreness, chest pain, nausea, vomiting, dizziness, and shortness of breath as mentioned in Table 7.

In domain four of Kidney Disease Quality of Life-36 contains information regarding the effects of kidney disease in daily life in patients undergoing maintenance haemodialysis. This study result indicates statistical significance (p value < 0.005) for the effects of kidney disease faced by CKD-ESRD patients as presented in Table 8.

4 Discussion

Chronic kidney disease and end-stage renal disease are widespread health concerns, with hemodialysis being a critical treatment option for sustaining life. Our study found that the majority of patients undergoing maintenance hemodialysis were between 51 and 60 years old, aligning with previous research indicating that individuals in this age group experience higher dialysis rates due to declining kidney function, comorbid conditions, and lifestyle factors [15]. Notably, male patients had a higher history of smoking, alcohol consumption, and tobacco use (91.1%), while only 33.3% of female patients reported tobacco use. A familial history of kidney disease was present in 21.8% of the cases, suggesting genetic predisposition.

Table 3 Cost of illness in study population underwent maintenance hemodialysis patients (N=55)

Cost of illness	Parameters	Cost (in INR)	Cost (in USD)	Cost basis	No of patients (N=55)	Percentage	Mean $\pm$ SD
Direct-medical cost	Hemodialysis session			Weekly			
	Two session	4000	Approx 48		13	23.6	4000 $\pm$ 0.000
	Three session	6000	Approx 48		42	76.3	6000 $\pm$ 0.000
	Laboratory investigation	100–200	1.20–2.40	Monthly	29	52.7	157.2 $\pm$ 23.93
		201–300	2.40–3.60		26	47.2	268.4 $\pm$ 27.39
	Injection						
	Erythropoietin	1300	16	Weekly	37	67.2	1300 $\pm$ 0.000
	Iron Sucrose	300	3.60		18	32.7	300 $\pm$ 0.000
	AV Fistula Access			One-time			
	Double vein	4550	55		30	54.5	4550 $\pm$ 0.00
	Triple vein	5950	72		25	45.4	5950 $\pm$ 0.00
	Medication	600–700	7.20–8.50	Monthly	16	29	645 $\pm$ 26.33
		701–800	8.50–9.60		11	20	751 $\pm$ 26.20
		801–900	9.60–10.80		13	23.6	857 $\pm$ 23.23
		901–1000	10.80–12.00		15	27.2	954 $\pm$ 28.95
Direct-non-medical cost	Food	100–200	1.20–2.40	Per visit	24	43.6	168.5 $\pm$ 23.17
		201–300	2.40–3.60		31	56.3	265 $\pm$ 25.70
	Transport	1–100	0.12–1.20	Per visit	29	45.4	47.6 $\pm$ 26.10
		101–200	1.20–2.40		26	54.4	164 $\pm$ 23.79
	Extra	100–300	1.20–3.60	Per visit	25	45.4	227.3 $\pm$ 55.93
		Above 300	Above 3.60		30	54.5	406.2 $\pm$ 55.54
Indirect cost	Wages	1000–3000	12–36	Per visit	28	50.9	2065 $\pm$ 474.4
		Above 3000	Above 36		27	49.0	4231 $\pm$ 503.3
	Caretaker	1000–3000	12–36	Per visit	26	47.2	2114.92 $\pm$ 594.2
	Wages	Above 3000	Above 36		29	52.7	4199.6 $\pm$ 565.12

SD: standard deviation; AV: Arteriovenous

Mean $\pm$ SD: These represent the average and standard deviation of costs, reflecting the variability in the data. It does not reflect the median cost but the mean (average) cost

Body mass index (BMI) analysis revealed that 65.5% of patients had a normal BMI, with 21.8% categorized as overweight and 9.1% as obese. Obesity is recognized as a risk factor for kidney disease due to the increased metabolic demands placed on the kidneys. Hemodialysis duration showed that 36.4% of patients had been undergoing treatment for one year, while 32.7% had been on dialysis for 1–2 years.

Our pharmacoeconomic analysis divided the cost of illness into direct costs (medical and non-medical), indirect costs (wages and caregiver wages), and intangible costs. Hemodialysis expenses, including equipment, medications, and laboratory tests, posed a significant financial burden, especially for patients from lower socioeconomic backgrounds. The average cost per session was INR 2000, (approximately USD 24) amounting to INR 192,000 (approximately USD 2304) annually for twice-weekly sessions, and INR 288,000 (approximately USD 3456) for thrice-weekly sessions [16, 17]. Patients with comorbid conditions required additional medications, further increasing the medical costs. These findings align with previous studies [18], which also highlighted the impact of comorbidities on treatment expenses.

Intangible costs were assessed using the Kidney Disease Quality of Life-36, which measures the physical and mental impact of kidney disease. In the first domain, patients experienced significant declines in both physical and mental functioning, consistent with previous research [19]. The burden of kidney disease (Domain 2) and symptoms during dialysis (Domain 3) were major factors contributing to a reduced quality of life, as observed in similar studies [20, 21]. In domain 4, patients reported significant lifestyle restrictions due to dietary and fluid intake limitations, reflecting the findings of prior studies on the daily impact of kidney disease [22].

Table 4 Standardized monthly and annual costs of illness for patients undergoing hemodialysis (N = 55)

Cost of illness		Cost (in INR)	Cost (in USD)	No of patients (N = 55)	Monthly cost (in INR) (in USD)		Annual cost (in INR) (in USD)	
Direct-medical cost	Hemodialysis session							
	Two session	4000	48	13	16,000	192	192,000	2304
	Three session	6000	72	42	24,000	288	288,000	3456
	Laboratory Investigation	100–200	1.20–2.40	29	157.2	1.89	1886.4	22.63
		201–300	2.40–3.60	26	268.4	3.24	3220.8	38.92
	Injection							
	Erythropoietin	1300	16	37	5200	64	62,400	768
	Iron Sucrose	300	3.60	18	1200	14.40	14,400	172.80
	AV Fistula Access							
	Double vein	4550	55	30	NA	NA	4550	55
	Triple vein	5950	72	25	NA	NA	5950	72
	Medication	600–700	7.20–8.50	16	645	7.74	7740	92.88
		701–800	8.50–9.60	11	751	9.01	9,012	108.14
		801–900	9.60–10.80	13	857	10.28	10,284	123.36
		901–1000	10.80–12.00	15	954	11.44	11,448	137.28
Direct-non-medical cost	Food	100–200	1.20–2.40	24	674	8.08	8,088	97.06
		201–300	2.40–3.60	31	1060	12.72	12,720	152.64
	Transport	1–100	0.12–1.20	29	190.4	2.28	2284.8	27.41
		101–200	1.20–2.40	26	656	7.87	7872	94.44
	Extra	100–300	1.20–3.60	25	227.3	2.73	2727.6	32.73
		Above 300	Above 3.60	30	406.2	4.88	4875	58.51
Indirect cost	Wages	1000–3000	12–36	28	8260	99.12	99,120	1189
		Above 3000	Above 36	27	16,924	203.09	203,088	2437
	Caretaker	1000–3000	12–36	26	8459.68	101.52	101,515	1218.24
	Wages	Above 3000	Above 36	29	16,798.4	201.58	201,580	2418.96

The costs for medical and non-medical components are standardized on a monthly and annual basis, Weekly costs for some components are multiplied by four to obtain monthly costs and by 52 to calculate annual costs. One-time costs such as AV fistula are not annualized, as they occur only once

Table 5 Domain-1 physical and mental functioning in haemodialysis patients

Domain 1	Physical and mental functioning	Mean ± SD	P value
1	Health	1.636 ± .8685	0.00**
2	Ability of moderate activity	1.455 ± .5025	
3	Ability of several activities	1.455 ± .5025	
4	Less work cause of physical health	1.491 ± .5045	
5	Kind of work	1.455 ± .5025	
6	Regular daily activities	2.691 ± 1.6983	
7	Difference in working ability	2.636 ± 1.0948	
8	Pain interfering normal work	2.909 ± 1.5064	
9	Felt calm	2.018 ± 1.3539	
10	Felt having lot of energy	4.273 ± 1.2696	
11	Felt downhearted	1.782 ± 1.3566	
12	Emotional problems interfering social activities	1.782 ± 1.1497	

**P < 0.05 considered statistically significant using Kruskal–Wallis test, SD: standard deviation

Table 6 Domain-2 burden of kidney disease in haemodialysis patients

Domain 2	Burden of kidney disease	Mean $\pm$ SD	P value
13	Disease condition interferes the life	1.655 $\pm$.9664	0.000**
14	Time spends in dealing with kidney disease	4.218 $\pm$ 1.2866	
15	Feeling frustrated in dealing with kidney disease	3.273 $\pm$ 1.6717	
16	Feeling burden to my family	1.509 $\pm$ 1.1201	

**P < 0.05 considered statistically significant using Kruskal–Wallis test, SD: standard deviation

Table 7 Domain-3 symptoms and problems in haemodialysis patients

Domain 3	Symptoms and problems subscale	Mean $\pm$ SD	P value
17	Muscle soreness	3.073 $\pm$ 1.6871	0.00**
18	Chest pain	2.164 $\pm$ 1.5488	
19	Cramps	2.200 $\pm$ 1.4957	
20	Itchy skin	3.309 $\pm$ 1.7200	
21	Dry skin	2.036 $\pm$ 1.4397	
22	Shortness of breath	2.109 $\pm$ 1.2718	
23	Faintness	3.182 $\pm$ 1.4918	
24	Lack of appetite	1.964 $\pm$ 1.4006	
25	Washed out	2.182 $\pm$ 1.6565	
26	Numbness in feet and hands	2.000 $\pm$ 1.4657	
27	Nausea And Stomach Upset	2.891 $\pm$ 1.4741	
28	Problem At Access Site	2.073 $\pm$ 1.2451	

**P < 0.05 considered statistically significant using Kruskal–Wallis test, SD: standard deviation

Table 8 Domain-4 effects of kidney disease on the daily life of haemodialysis patients

Domain 4	Effects of kidney disease on daily life subscale	Mean $\pm$ SD	P value
29	Fluid restriction	1.745 $\pm$ 1.1258	0.00**
30	Dietary restriction	1.564 $\pm$ 1.1016	
31	Ability to do work around the house	1.327 $\pm$.9038	
32	Ability to travel	2.273 $\pm$ 1.3670	
33	Depending health care professionals	1.091 $\pm$.3482	
34	Stress	3.855 $\pm$ 1.5082	
35	Sex life	1.582 $\pm$.7623	
36	Personal appearance	2.636 $\pm$ 1.7571	

**P < 0.05 considered statistically significant using Kruskal–Wallis test, SD: standard deviation

Direct medical costs, driven by hemodialysis-related expenses, were higher than non-medical and indirect costs. Expanding the availability of public hemodialysis facilities could reduce these costs [18]. Patients also faced intangible costs, including the emotional and physical toll of treatment, which significantly diminished their quality of life.

4.1 Strengths and limitations

This study is one of the few to examine the economic burden of hemodialysis in rural areas, offering insights that can guide policy development. The inclusion of diverse rural participants enhances the generalizability of the findings. However, limitations include potential underestimation of the disease burden due to the use of quality-adjusted life years (QALYs) and the time horizon selected for cost data analysis. Addressing these limitations can improve the accuracy of future pharmacoeconomic studies [23, 24].

4.2 Impact on policy, research and clinical application

Expanding access to government-funded hemodialysis centers in rural areas is crucial. Increased research funding is necessary to develop better treatments and reduce the financial burden on patients. In clinical practice, patient-centered care, shared decision-making, and improved patient education are essential for managing Chronic kidney disease end-stage renal disease effectively. Auditing hospital-based issues such as outdated equipment and Chronic kidney disease end-stage renal disease patients in rural areas. To address this, policy must expand rural hemodialysis centres, address resource constraints, and raise kidney disease awareness, and clinical care must prioritize patients, engage them in decision-making, and ensure safe hospital practices.

5 Conclusion

The cost of maintenance haemodialysis significantly burdens patients and caregivers, particularly those from lower socioeconomic backgrounds in rural areas. This study emphasizes the need for improved access to hemodialysis treatment and support services for Chronic kidney disease–End-stage renal disease patients in rural areas. To address this, policy must expand rural hemodialysis centers, address resource constraints, and raise kidney disease awareness, and clinical care must prioritize patients, engage them in decision-making, and ensure safe hospital practices.

Acknowledgements We are very grateful to acknowledge our Head of the Institution Dr. G. Murugananthan, M.Pharm., Ph.D., and Dr. P. Sharmila Nirojini, M.Pharm., Ph.D., Head, Dept of pharmacy Practice, Swamy Vivekanandha College of Pharmacy & Faculty Advisor of ISPOR India Tamilnadu Regional Chapter for their valuable guidance and suggestions for this research.

Author contributions Nithya Raju contributed to the study design, drafting of the manuscript and critical review. Redlin Jani and Reema Mohammed Sultan contributed to the drafting of the manuscript. Sanjana Mariam Saju and Paneerselvam Theivendren contributed to the data acquisition, analysis and critical review. All authors read and approved the final manuscript.

Funding None.

Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request. All data generated or analyzed during this study are included in this published article Additional data related to this study are available in the supplementary materials.

Declarations

Ethics approval Ethical approval was obtained from the Institute Ethics committee of Vivekanandha Medical care Hospital, File Ref. No: EC/NEW/INIT/2021/1811 reviewed and discussed to conduct the study. (Ref. no: SVCP/IEC/SEP/2021/009).

Consent to participate An informed consent was obtained from the individuals on maintenance hemodialysis who wanted to participate in this study before they filled in the questionnaire.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Varughese S, Abraham G. Chronic kidney disease in India: a clarion call for change. *Clin J Am Soc Nephrol*. 2018;13(5):802.
2. Williams ME. Diabetic CKD/ESRD 2010: a progress report? *Seminars Dial*. 2010;23:129–33.

3. Agarwal S, Srivastava R. Chronic kidney disease in India: challenges and solutions. *Nephron Clin Pract.* 2009;111(3):c197–203.
4. Teerawattananon Y, Luz A, Pilasant S, Tangsathitkulchai S, Chootipongchaivat S, Tritasavit N, et al. How to meet the demand for good quality renal dialysis as part of universal health coverage in resource-limited settings? *Health Res Policy Syst.* 2016;14(1):1–8.
5. Veerappan I, Arvind R, Ilayabharthi V. Predictors of quality of life of hemodialysis patients in India. *Indian J Nephrol.* 2012;22(1):18.
6. Rosansky SJ, Schell J, Shega J, Scherer J, Jacobs L, Couchoud C, et al. Treatment decisions for older adults with advanced chronic kidney disease. *BMC Nephrol.* 2017;18:1–10.
7. Bharati J, Jha V. Global dialysis perspective: India. *Kidney360.* 2020;1(10):1143.
8. Kumar R. Medical tourism in India (management and promotion). New Delhi: Deep and Deep Publications; 2008.
9. Marten R, McIntyre D, Travassos C, Shishkin S, Longde W, Reddy S, et al. An assessment of progress towards universal health coverage in Brazil, Russia, India, China, and South Africa (BRICS). *The Lancet.* 2014;384(9960):2164–71.
10. Berman P, Ahuja R. Government health spending in India. *Economic and Political Weekly.* 2008:209–16.
11. Sahoo PM, Rout HS, Jakovljevic M. Future health expenditure in the BRICS countries: a forecasting analysis for 2035. *Glob Health.* 2023;19(1):1–17.
12. Jha V. Current status of end-stage renal disease care in South Asia. *Ethn Dis.* 2009;19:27–32.
13. Rao M, Juneja R, Shirley R, Jacob CK. Haemodialysis for end-stage renal disease in Southern India—a perspective from a tertiary referral care centre. *Nephrol Dial.* 1998;13(10):2494–500.
14. Sharma R. Revision of Prasad's social classification and provision of an online tool for real-time updating. *South Asian J Cancer.* 2013;2(3):157.
15. Tong A, Evangelidis N, Kurnikowski A, Lewandowski M, Bretschneider P, Oberbauer R, et al. Nephrologists' perspectives on gender disparities in CKD and dialysis. *Kidney Int Rep.* 2022;7(3):424–35.
16. Suja A, Anju R, Anju V, Neethu J, Peeyush P, Saraswathy R. Economic evaluation of end stage renal disease patients undergoing hemodialysis. *J Pharm Bioallied Sci.* 2012;4(2):107.
17. Wang V, Vilme H, Maciejewski ML, Boulware LE. The economic burden of chronic kidney disease and end-stage renal disease. *Seminars Nephrol.* 2016;36:319–30.
18. Ahlawat R, Tiwari P, D'Cruz S. Direct cost for treating chronic kidney disease at an outpatient setting of a tertiary hospital: evidence from a cross-sectional study. *Value Health Regional Issues.* 2017;12:36–40.
19. Filipčič T, Bogataj Š, Pajek J, Pajek M. Physical activity and quality of life in hemodialysis patients and healthy controls: a cross-sectional study. *Int J Environ Res Public Health.* 2021;18(4):1978.
20. Gunarathne TGNS, Tang LY, Lim SK, Nanayakkara N, Damayanthi HDWT, Abdullah KL. Factors associated with symptom burden in adults with chronic kidney disease undergoing hemodialysis: a prospective study. *Int J Environ Res Public Health.* 2022;19(9):5540.
21. Cox KJ, Parshall MB, Hernandez SH, Parvez SZ, Unruh ML. Symptoms among patients receiving in-center hemodialysis: a qualitative study. *Hemodial Int.* 2017;21(4):524–33.
22. van Haalen H, Jackson J, Spinowitz B, Milligan G, Moon R. Impact of chronic kidney disease and anemia on health-related quality of life and work productivity: analysis of multinational real-world data. *BMC Nephrol.* 2020;21(1):1–15.
23. Moradi N, Rashidian A, Nosratnejad S, Olyaeemanesh A, Zanganeh M, Zarei L. The worth of a quality-adjusted life-year in patients with diabetes: an investigation study using a willingness-to-pay method. *Pharmacoeconomics-open.* 2019;3:311–9.
24. Jo C. Cost-of-illness studies: concepts, scopes, and methods. *Clin Mol Hepatol.* 2014;20(4):327.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.